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Experimental investigation of microvibrations induced by reaction wheels on earth observation satellite

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Abstract

The evaluation of microvibrations effect is crucial for the success of remote sensing missions. The moving parts on the satellite such as reaction wheels can create microvibrations disturbances, which cause degradation of image quality. In our paper we present an experimental investigation methodology for on-ground quantification of wheel noise disturbance transmitted through the structure to the payload. This assessment was carried out for the **ALSAT-1B** satellite integrating a medium resolution payload. The angular motion along imager bore sight was measured using accelerometers and verified against image quality and mission requirement. The on ground experimental methodology suggests that the effect of microvibrations on a camera can be verified directly.

Keywords: Microvibration test; Payload; Fundamental frequency; Reaction wheel; Image motion

1. Introduction

In the last decade, high resolution satellites are progressing rapidly. One of the main mission concerns is the impact of pointing stability on image resolution. An important issue can effect this requirement is microvibrations sensitivity of instruments generated from the moving parts onboard the satellite, which must be mitigated effectively in order to obtain high pointing accuracy (Cobb et al., 1999).

Reaction wheels (RWA) are an active mechanism used for altitude control which gives the required pointing accuracy for high resolution spacecraft (Kamesh et al., 2010). They are becoming particularly important in the modern space industry, due to increasing demands for better maneuverability, agility and precision control of small satellites. However, due to static imbalance, dynamic imbalance and bearing imperfections, vibrations are generated when hard-mounted on an isolated and rigid platform (Lee et al., 2015; He et al., 2020). The microvibrations produced by reaction wheels are transferred through the satellite structure which excites dynamic modes of the imager and can limit its performance (Toyoshima et al., 2003). Consequently, the produced disturbance affects pointing accuracy and imaging quality (Nagabhushan and Fitz-Coy, 2014; Yang et al., 2013; Kim, 2014; Liu and Luo, 2016). The Fig. 1 represents an example of image distortion due to microvibration disturbance.

Micro-vibration test is necessary to predict and verify the observation performance of the spacecraft on orbit. Hence, the understanding and control of the vibration level at sensitive locations, using passive damping and active control technologies is also a crucial factor in order to achieve the desired instruments' performance (Vaillon et al., 2002; Bae et al., 2012; He et al., 2018; Mankour, 2020). The assessment of microvibrations impact on image quality is a complex process which includes structure,

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Table 1
Location of accelerometers and channel of data acquisition.

Ch. N°	Description	Location
1	Ambient noise measurement	101
2	Imager lower mount plate on satellite Y axis	11Y
3	Imager Lens Mount on satellite Y axis	12Y
4	Imager lower mount plate on satellite X axis	13X
5	Imager lens mount on satellite X axis	14X
6	In the centre of bolts attachment of Y wheel	22Z
7	In the centre of bolts attachment of Z wheel	23Z
8	In the centre of bolts pattern of X wheel	24Z

Fig. 1. Microvibration effect on geometric distortion (Zhang et al., 2011a).

control and optics (Toyoshima et al., 2010; Kawak, 2017). Therefore, the quantification of microvibrations is hard to obtain due to low level vibration compared with the dynamic environment during launch. The test method and the test system are different than the testing of usual launch dynamic environment (Roshbe et al., 2013).

Two test methods are adopted in the literature to assess the microvibrations issues for optical satellites. The first approach relies on an optical imaging test, which simulates the camera in orbit to assess the influence of microvibrations (Li et al., 2018; Wang et al., 2018). Then, the perturbation effect on the image can be measured by the optical testing (Bronowicki, 2006; Zhou et al., 2011). The second approach reposes on the use of high-precision accelerometers to measure the mechanical response characteristics of the desired components and, derive the overall system performance.

On ground tests can be achieved based on high-precision accelerometers for microvibration evaluation. Therefore, the measurement is categorized into two sub-groups with respect to uncoupled and coupled boundary condition. The uncoupled measurements are used only for the prediction of spacecraft pointing stability (Shankar Narayan et al., 2008; Vitelli, et al., 2014). Zhang et al., 2011b, showed that RWA induced lateral and axial microvibrations which are accurately measured using a seismic-mass microvibrations measurement system. Addari et al.2017, studied a method involves measurements of the disturbances induced by reaction wheel assembly in a hard-mounted configuration and speed dependent dynamic mass of the reaction wheel. Zhou et al., 2012, conducted an experimental microvibration tests produced by MWA by analyzing the disturbance sources created by mass imbalance, structural mode, bearing irregularity and nonlinear stiffness, and random noise. While, coupled measurement method is performed at flight spacecraft level (Laurens, et al., 1996), which is considered in our analysis. Consequently, we can measure microvibrations along the transmission path as well as optical system and verify the image shift. Moreover, ground testing validation is important to calibrate and checked the accuracy of the numerical model calculations (Park et al., 2014).

Other researcher achieved in-orbit measurement of microvibrations onboard a remote-sensing satellite related noise level, disturbance source features, and satellite structure transmission characteristics (Ooi et al., 2007; Dengyun et al., 2019). Measurements in orbit contain the vibratory signals from all other sources. Hence, this interference needs to be treated with extreme care. In addition, the in orbit measurements require dedicated equipment and sensors which lead to an increase of the spacecraft cost.

Several researchers had adopted the angular motion measurement dedicated for the evaluation of the pointing stability performance of the sensitive payload in the microvibration test and fuzzy (Zhou et al., 2012; Hu et al., 2014). The angular displacements of the optical payload was measured in-ground and in-orbit based on the high-precision acceleration sensor showed high consistency with the results of the imaging test (Chen et al., 2019). The analysis of flywheel microvibration influence on the camera was determined regarding the largest angular displacement of the secondary mirror along the optical axis direction (Li et al., 2018). The microvibration of different directions causes corresponding translational or rotational displacement, and eventually produces image motion on camera optical plane (Zhang, 2013; Xue et al., 2016). The evaluation and prediction of image quality are important tasks especially when microvibrations are considered (Toyoshima et al., 2003; Lee, et al., 2012a).

This paper presents on ground experimental methodology to measure and assess microvibrations levels transmitted from RWA to the imager, which can affect the image quality of a medium resolution payload of ALSAT-1B satellite. The angular motion along the imager axis was realized by the quantification of microvibrations transferred from the three spinning wheels to the payload based on the worst case scenario during imaging. Therefore, the image quality was verified regarding the imager angular motion.

2. Materials and methods

The Alsat-1B satellite is shown in Fig. 2, which is an Algerian satellite for agricultural and disaster monitoring, launched into a 670 km sun-synchronous orbit on board the PSLV launch vehicle from the Sriharikota in India on 26 September 2016. The satellite is based on the SSTL-100 bus. The satellite weighs is 103 kg and carries an earth imaging payload with 12-metre panchromatic imager and 24-metre multispectral cameras delivering images with a swath width of 140 km.

The physical dimensions of ALSAT-1B cubic form are as follow: Width in X = 640 mm, Width Y = 765 mm, Height in Z = 1090 mm. While, attitude actuation is insured by using three 10SP-M reaction wheels (Fig. 3) arranged in Y, X and Z directions (Fig. 4). The reaction wheels are 109 mm in diameter and 101 mm long with a mass of about 960 g using integrated electronics that execute commands from the onboard computer. The reaction wheels spin at up to 5000 RPM to deliver an angular momentum of 0.42 Nms and a peak torque of 0.011 Nm.

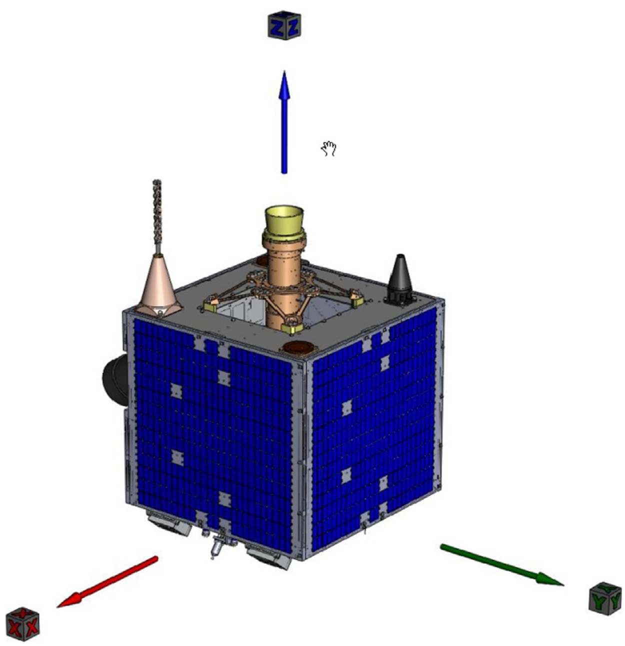

Fig. 2. ALSAT-1B External view.


Fig. 3. 10 SP-M wheel.

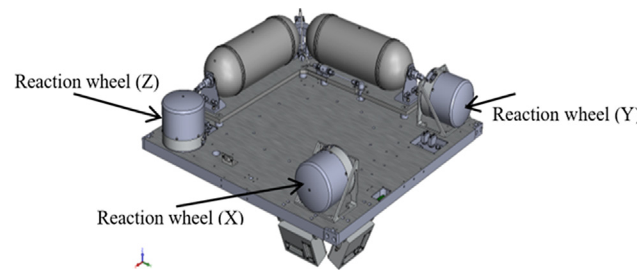

Fig. 4. Reaction wheels configuration.

Preliminary analysis showed that, the disturbance from hard mounted 10-SP M reaction wheels could potentially cause imaging quality degradation. Hence, experimental tests were needed to assess the imager angular motion levels caused by microvibrations disturbance. We note that, the satellite embark three wheels towards the satellite axis (X, Y and Z) for the stabilization purpose (Fig. 4).

2.1. Experimental procedure and testing plan

A set of testing has been undertaken on the Proto-Flight Model (PFM) spacecraft to assess the microvibrations effect on ALSAT-1B imager. Due to very low response levels, an extremely quiet test environment was needed for a successful test. Consequently, wheels disturbance testing was performed during night and air conditioning in the clean room was switched off to reduce the maximum testing environment noise.

2.2. Test configuration

The spacecraft was suspended vertically supported by an isolation frame (Fig. 5). The isolation frame serves two purposes for this testing. Firstly, isolates the satellite from any ambient vibration coming through the floor of the building, and secondly, permits boundary conditions to be more closely mimic as on orbit conditions. Thus, the test is performed under free-free conditions.

- **Accelerometers positions**

For disturbance noise measurement on the imager, four (04) piezoelectric accelerometers (PCB 333B50) were attached on the imager. All accelerometers were mounted to the outside of imager structure (Fig. 6); two (02) were mounted close to the focal plane assembly (FPA) labeled as 11Y and 13X and, and two (02) near the lens labeled as 12Y and 14X. We note that, all accelerometers were single axis measuring the radial acceleration pointing towards the center axis of the imager. Moreover, all accelerometers direction should align with S/C axis.

In addition, the measurement of microvibration levels produced of each wheel was achieved by using three accelerometers (03) attached on the supporting plate (22Z, 23Z and 24Z). The accelerometers are located at

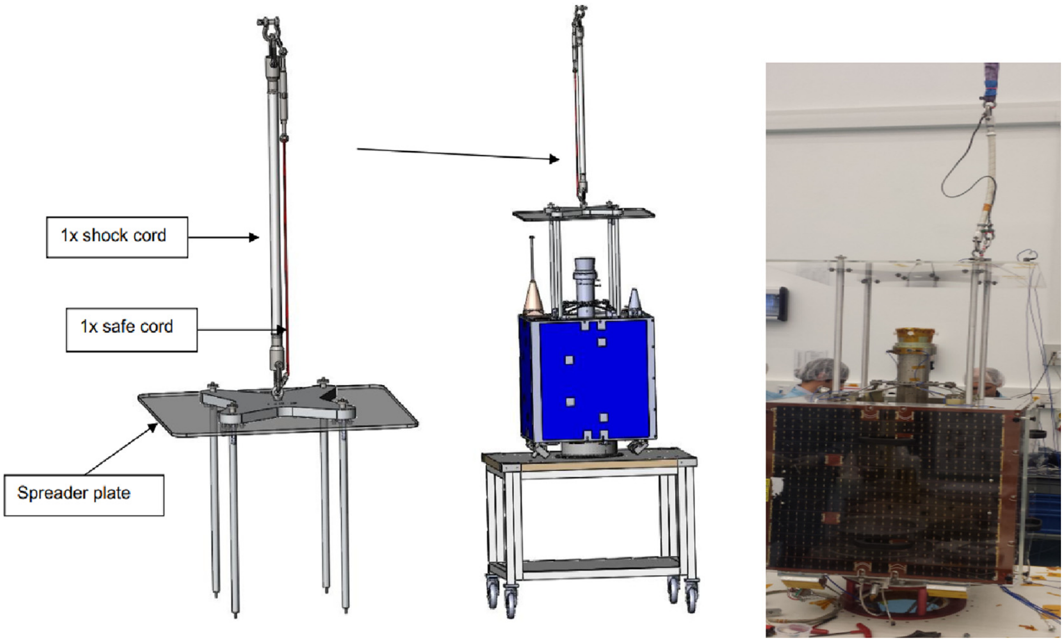

Fig. 5. Microvibrations Test Setup.

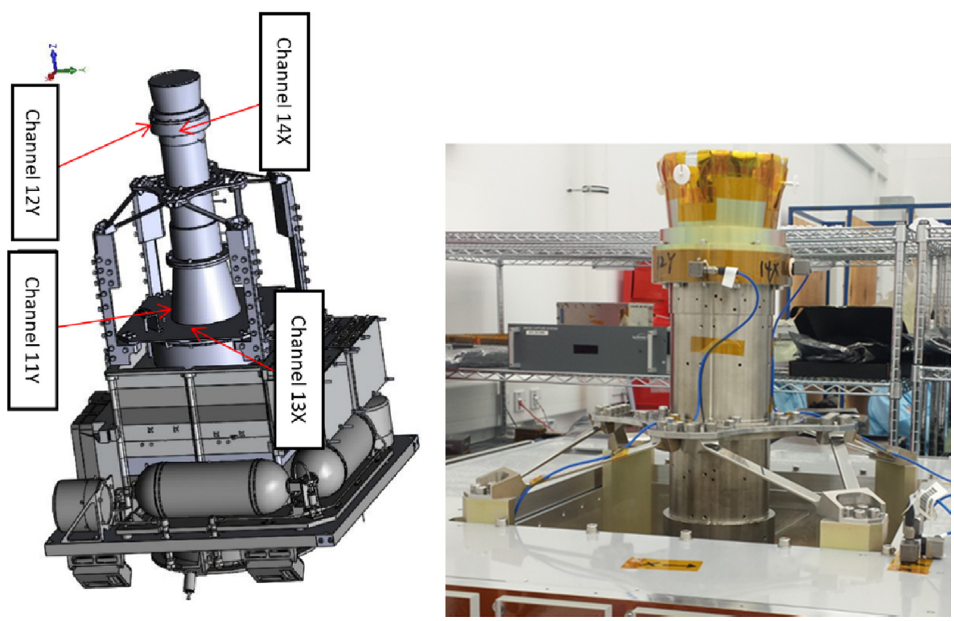

Fig. 6. Accelerometers locations on the imager.

the middle of the reaction wheel bolt pattern (Fig. 7). This locations permit to measure the equivalent disturbance produced by each spinning wheel and transferred to the supporting plate. Therefore, an extra accelerometer was used (1 0 1) to measure the ambient noise to check the correctness of test results.

The following table details the data acquisition channels numbers and its respective locations used on the spacecraft during noise disturbance tests (Table 1).

2.3. Test setup and procedure

To evaluate the imager behaviour, some tests were carried out on the flight spacecraft with reaction wheels running through a defined speed profile dedicated for ALSAT-1B mission, which embark a medium resolution payload. While, testing with the reaction wheel as a disturbance source is useful for assessing the overall response of the spacecraft imager to realistic vibration input.

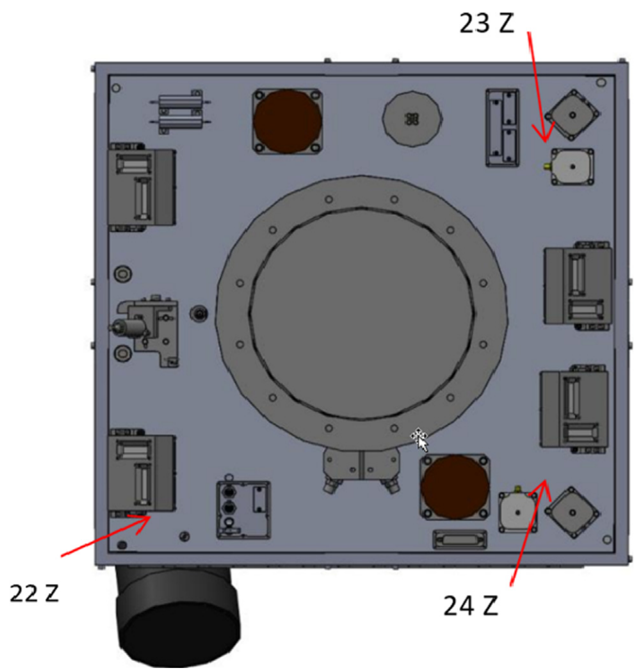

Fig. 7. Accelerometers positions on wheels supporting plate.

However, the performed tests do not represent the real in-orbit environment due to self-weight on wheels changing the bearing preload and differences in S/C damping due to the presence of air (Privat, 1999). Consequently, the microvibration disturbance level in orbit is higher than on ground (Toyoshima et al., 2010, Privat, 1999). Then, the results are scaled by a safety factor (SF) representing the difference of between in-ground and in-orbit microvibrations levels data deduced from a previous mission identified by the TDS-1 satellite wheel data. The data needs to be factored during the prost-processing to evaluate the potential worst case disturbance level on the imager in orbit.

The assumption to calculate the scaling factor that, in ground each instrument excites the same vibration disturbance as in orbit. Based on this assumption, the ground test data can be used in orbit regarding vibration level at the point of disturbance sources. Moreover, in order to make this assumption more accurate for our evaluation, we use the ground test data which were taken under the same RWs' operational condition as in orbit. To account for the difference in the behaviour regarding wheel speed variation, the scaling factor is calculated using the ratio of the maximum response over a range of wheel speeds on orbit divided by the maximum response over a range of wheel speeds on ground. Hence, this gives a ratio for each accelerometer channel. Therefore, the average of accelerometers ratios is taken to provide the final scaling factor. A factor of 18 has been agreed taking into account TDS-1 real data. As well as the ground to in-orbit data factor, another factor of 1.5 will need to be applied (Komatsu and Uchida, 2014) to account for differences between microvibrations test campaign in AIT and flight configurations and wheels characteristics degradation and variation in noise behavior over life. A scale factor "SF = 27" has been agreed to be used on top of the microvibrations noise.

The spacecraft during the test was set according to the configuration shown in Fig. 5, and the power system was turned on to rotate the three wheels. The details of wheels speed used during the tests are shown in Fig. 8.

3. Results and discussions

3.1. Microvibrations levels

The microvibration levels produced by each wheel (X, Y and Z) where determined according to mission profile scenario of wheels speed. The results of 0.2 s are extracted from the channels 22Z, 23Z and 24Z records (Fig. 9) during

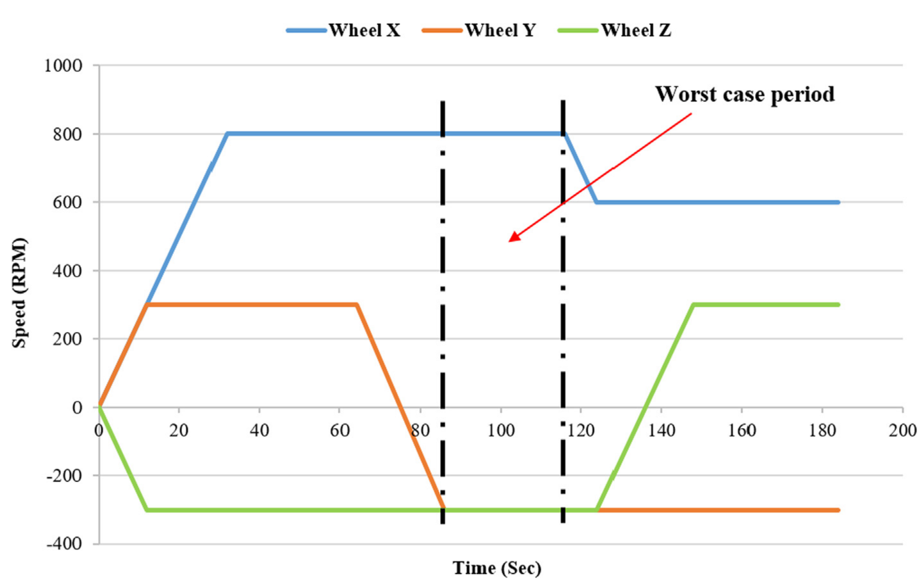

Fig. 8. The profile of wheel speed for microvibrations testing.

of the worst case period. This data shows that, the Y wheel is dominant in the microvibrations disturbance (2.57 g). Hence, Y wheel will be the main source of noise for the platform running with 800 rpm and X and Z wheel will be run at a max of 300 rpm. Therefore, higher rotation speed engenders more disturbance noise. Since, the RWA vibration amplitude is proportional to the wheel speed (Inamori, et al. 2013; Wang et al., 2018; Li et al., 2018).

3.2. Imager distortion

The microvibration levels on the imager were recorded through the channels 11X, 12X, 13Y and 14Y illustrated in the Figs. 10 and 11. The peak value (0.78 g) was recoded by the channel (14X). We remark that, the level of the microvibrations are attenuated compared to the disturbance source mounting positions (22Z, 23Z and 24Z), and it is remarkably reduced through the structure transmission (Bronowicki, 2006; Minghui et al., 2021). The signal to noise ratio was satisfactory so the assessment of the microvibration performance was based directly on these tests.

The recorded accelerations on the imager permit to determine the angular motions of the imager bore sight. The angular motion was calculated between the lens and FPA (Fig. 12) by processing MATLAB program. The

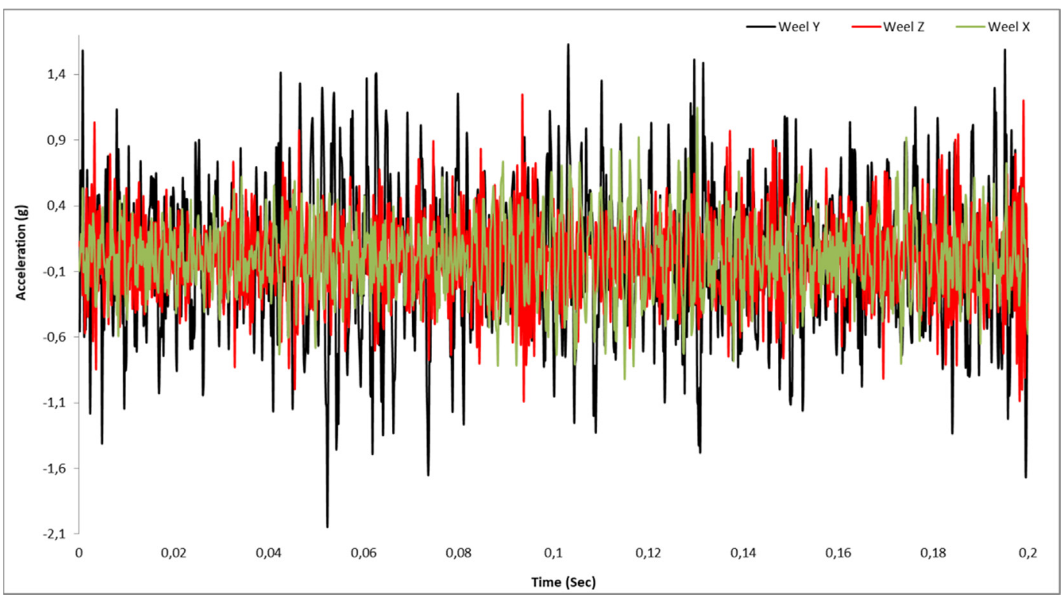

Fig. 9. Microvibrations vs. time variation for each wheel.

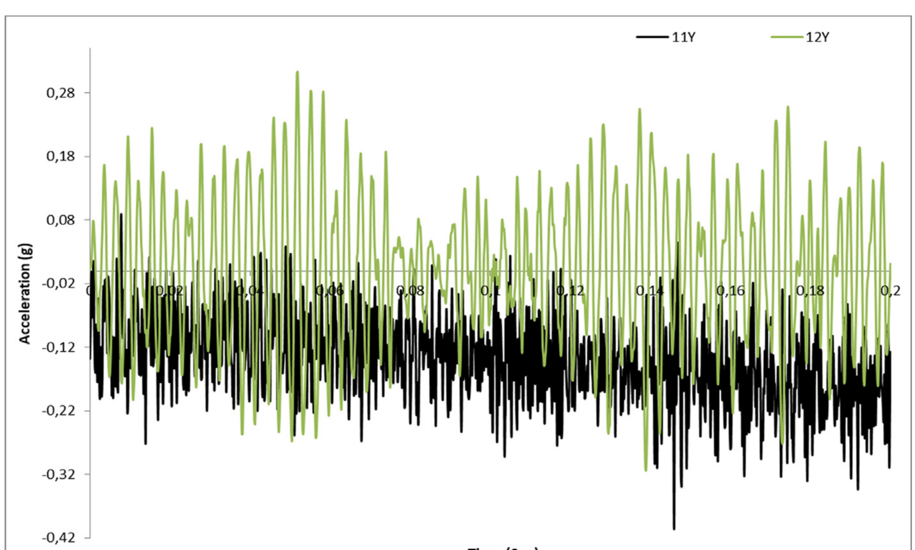

Fig. 10. Microvibrations levels vs. Time on the imager (13y and 14y).

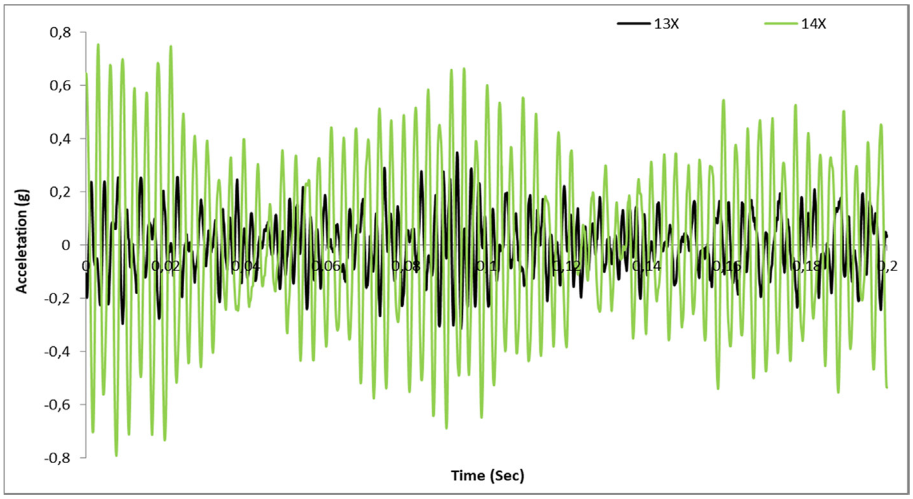

Fig. 11. Microvibrations levels vs. time on the imager (13X and 14x).

distance between the mounting plate (FPA) and the lens is 43 mm. microvibration mainly causes the optic axis to produce the lateral deviation. The translation along the axial direction has no effect on system imaging. The Figs. 13–18 show the imager angular motions in time and frequency domains towards X and Y directions. Therefore, all data is extracted from a 0.2 s during the worst case period (Fig. 8). We note that, the interested frequency band was up to 500 Hz.

Looking at the frequency domain plots of microvibrations responses along X and Y axis, it can be seen that, the results are broadly similar (Fig. 18) due to the symmetrical geometry of the satellite primary structure. Consequently, a peak amplitude value of order 2.2 10^{-4} Rad (frequency domain) appear around 140 Hz in both axes, which represents the fundamental frequency of the imager in each direction. According to modal analysis illustrated on the Fig. 19, the first resonance frequency of the imager is around 140 Hz and the second is around 350 Hz. Hence, the frequency of the induced disturbances matches the natural frequencies of the imager (Lee et al., 2012b). The comparison of angular displacement along optical payload axis show that, the larger value of imager focal plane shift was found 3.83 e^{-7} Rad in the X direction (Fig. 17). We can note also that, the results in X direction are slightly higher than in the Y direction by a difference of 6% (Fig. 17). For the comparison of experimental results, we repose on the work of **Cui et al.**, which found that the angular displacement of the camera is about 3.1 e^{-7} Rad. We note that, the structural attenuation of microvibration is different from satellite to another.

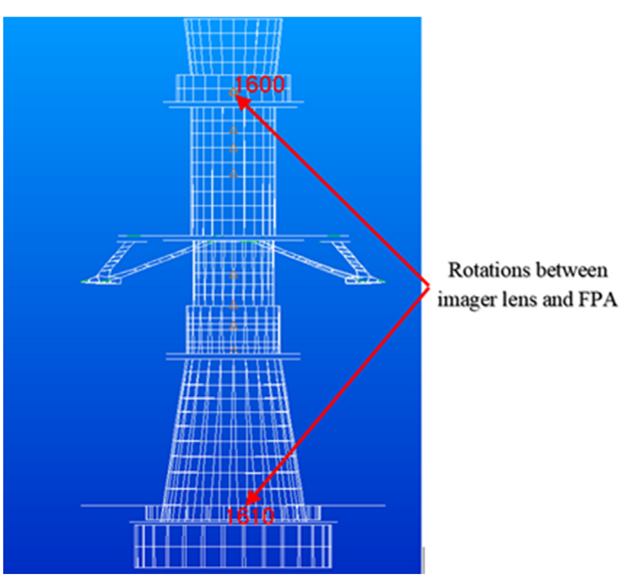

Fig. 12. Locations of lens and FPA of the imager.

To assess image distortion by processing the MATLAB program, we had added the angular displacement towards X axis obtained during worst case period scaled by 27 times to the image. Hence, the image offset can be calculated. Thus, the applicability of the spacecraft in microvibration environment is verified. Therefore, microvibrations levels are not expected to cause distortion in the image (Fig. 20.b). On the other hand, higher microvibrations levels (SF = 50) induced noticeable image distortion (Fig. 20.c).

With this set of testing completed and after checking the results, it was possible to show that, the microvibration performance of the ALSAT-1B spacecraft is adequate to fulfill its mission objectives. Although, the Fig. 21 taken in orbit demonstrates that, the microvibration disturbance levels produced by three wheels don't affect the image quality.

On ground experiment using flight model satellite is an effective methodology to a direct assessment in-orbit microvibration effect on image quality. Also, it gives real disturbances levels of spinning wheels closely similar to in orbit scenario. Hence, the ability to mitigate any issues leading to image degradation before launch. Whereas, other researchers repose on in-obit test to assess the impact

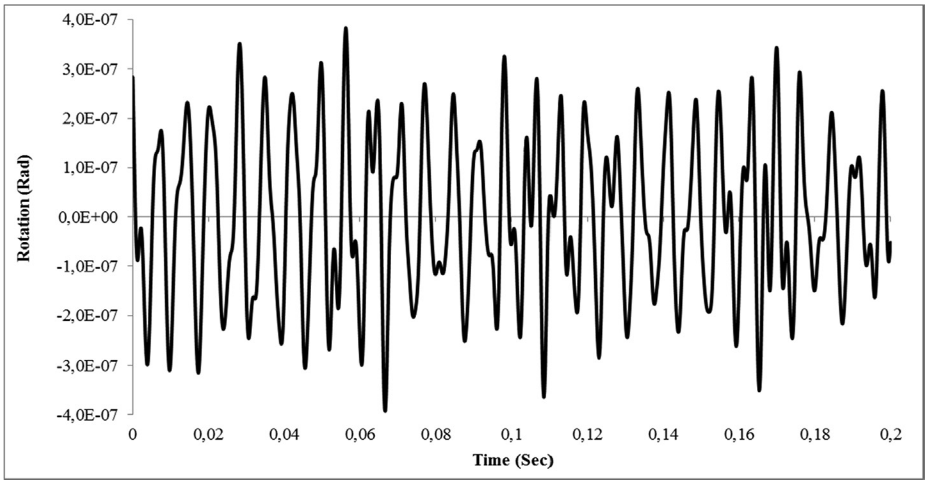

Fig. 13. Rotations vs. time variation of the imager towards X Axis.

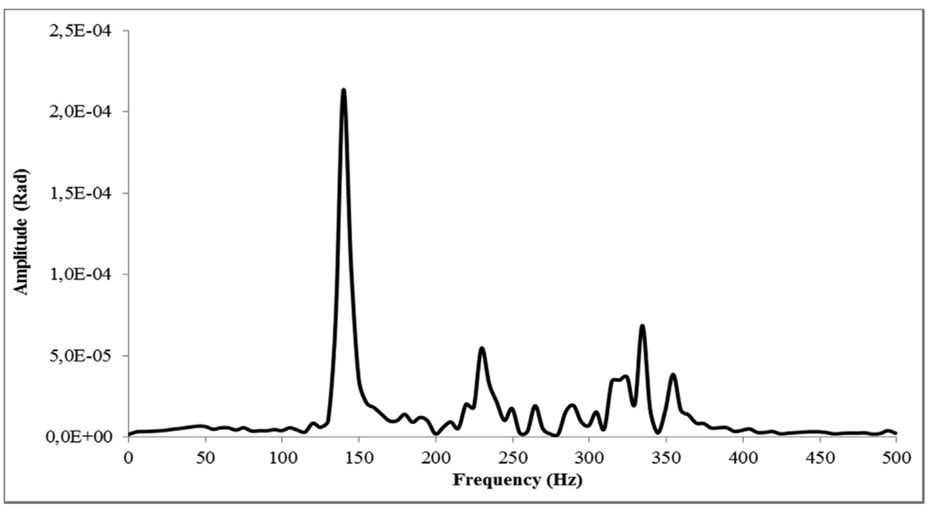

Fig. 14. Amplitude vs. frequency variation of the imager towards X direction.

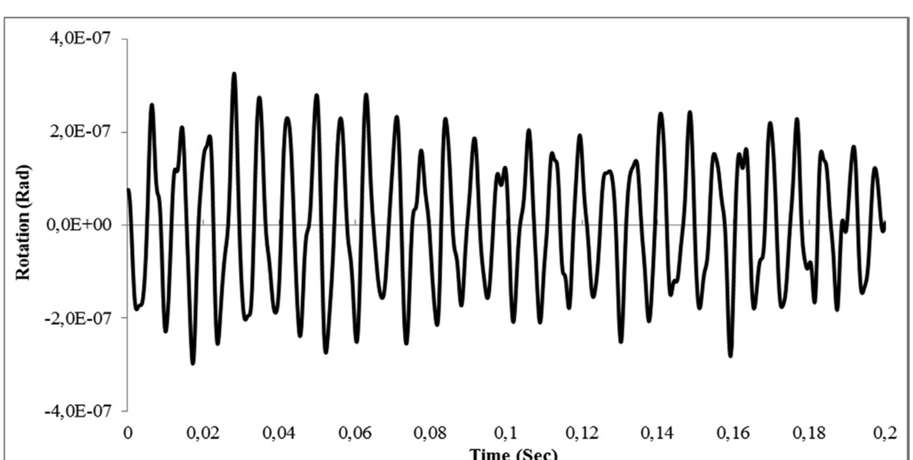

Fig. 15. Rotations vs. time variation of the imager towards Y Axis.

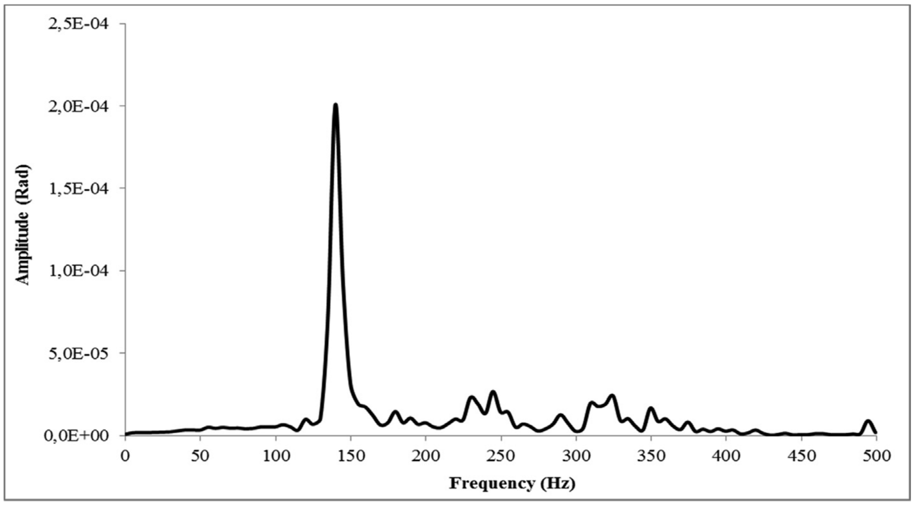

Fig. 16. Amplitude vs. frequency variation of the imager towards Y direction.

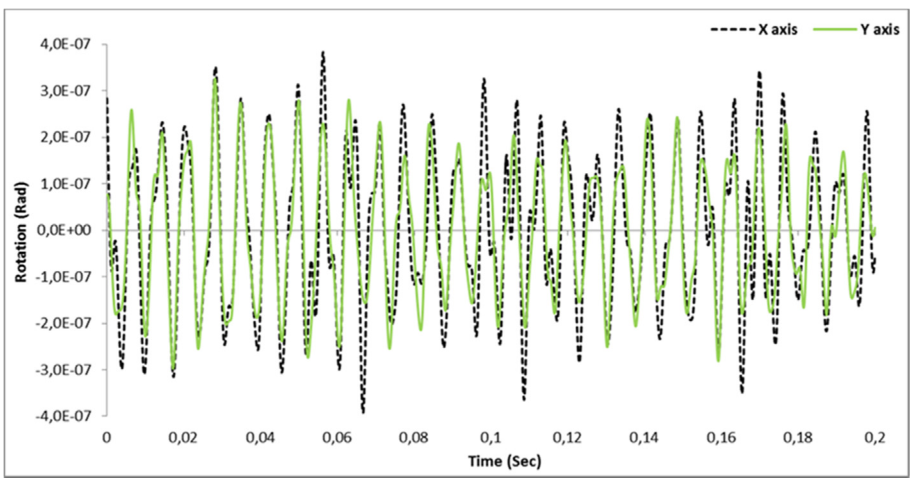

Fig. 17. Comparison of the imager Rotation vs. Time.

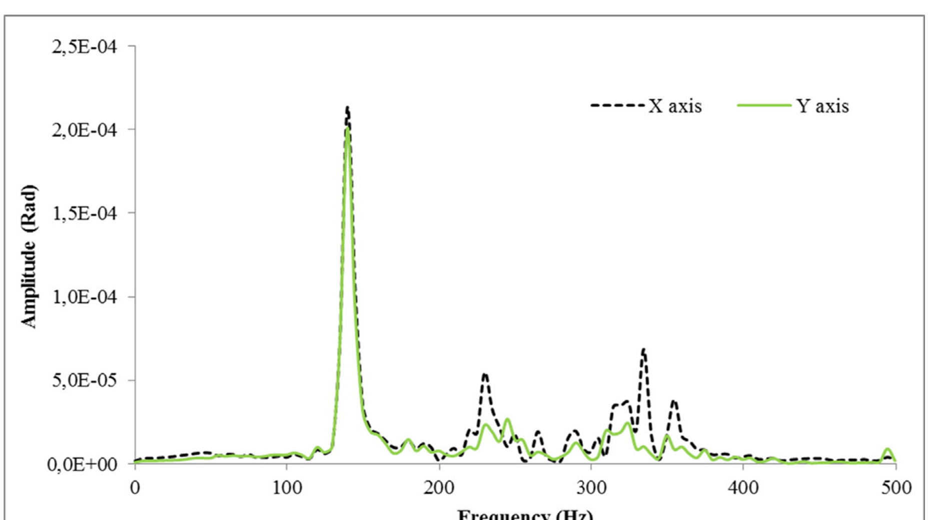

Fig. 18. Comparison of the imager Amplitude vs. Frequency.

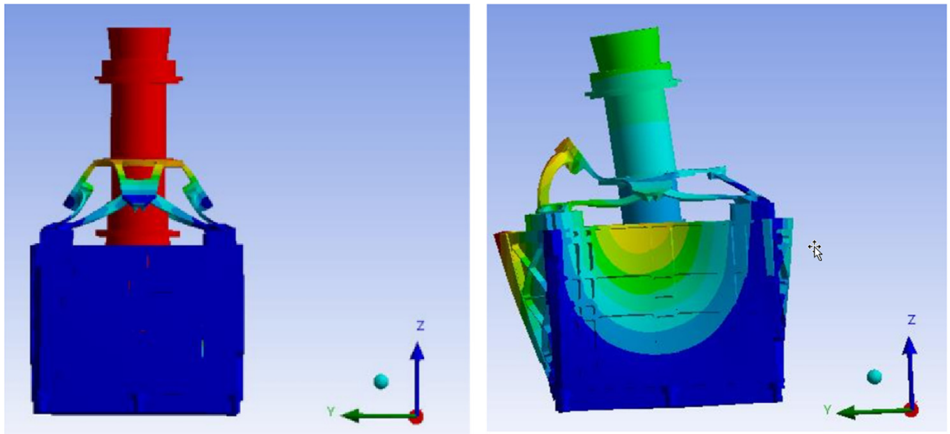

Fig. 19. Imager modes.

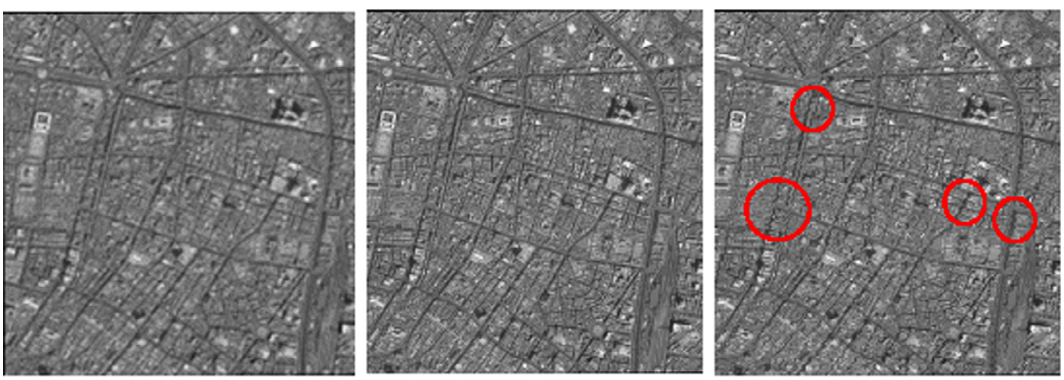

Fig. 20. Microvibrations effect on the image.

Fig. 21. Image taken by ALSAT-1B satellite.

of micro-vibration on the space camera and verify angular motion of optical payload and image quality (Cui et al., 2019; Dengyun et al., 2019).

4. Conclusion

The on ground evaluation of microvibration disturbance is a crucial activity to manage potential causes of image degradation. The experimental test allows a direct method to determine the amount of imager distortion-caused by wheel disturbance noise transferred to the payload through the satellite structure. Thus, test data have been used to calibrate the numerical model which provided the final verification. In addition, ground results should be factored to obtain the reel disturbance levels in orbit. Finally, the verification of image motion caused by the microvibration disturbance was obtained. For our imager case, the results are satisfactory to meet the mission objectives. Other microvibration sources can impact the image quality as the sloshing phenomenon (Marsell et al., 2009; Bourdelle et al., 2019), which can be more important when the fuel level decrease in the tank. Also, the integration of high resolution imager in this platform may lead to image quality deterioration; especially, the need of increasing wheels spun for better satellite pointing stability.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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